Electronics Workshop

How to teach this class — what happens, how to run it, and why it matters. The day the team learns to *read* electricity, before it wires the robot.

THE ONE GOAL

Every kid can measure a circuit with the meter, read the schematic symbols, and build a working circuit — the language of electronics, before the control box.

HOW THIS CLASS IS TAUGHT — MATE TEACHES, YOU FACILITATE

Assign MATE's electronics presentations as **pre-watch homework**; meeting time goes to measuring and building. One three-hour session. Source: the TriggerFish ROV Guide.

BEFORE YOU TEACH THIS

- **Watch the pre-watch yourself** — MATE's Multimeter Basics, How Switches Work, Simple Circuits.
- **Use a multimeter yourself** — measure a battery's voltage and beep out continuity, so you can coach it.
- **Print the Electrical Symbols handout** — one per student, three-hole punched for the notebook.
- **Read the assigned pages** — *Underwater Robotics* Ch 8 §4.4 (Ohm's Law) and §4.8 (open/short circuits). The grounding below.
- **Confirm the Simple Circuits Lab Kits arrived** — one per group, with a 9 V battery each.
- **Section 4.3:** every kid sets the meter, takes the readings, and wires their own circuit. Hands off.

WHY THIS CLASS MATTERS

Before the team wires the actual robot, it has to learn to read electricity, not just connect it. The multimeter turns guessing into knowing — is the path complete, what's the voltage, where's the open? The simple-circuits lab makes a switch, a battery, and a light into a concept the kids own. And the schematic symbols are the language every wiring diagram — and the competition SID — is written in. Get this class right and the control box build that follows is calm instead of chaotic.

THE ELECTRICAL "WHY" — FROM THE ASSIGNED READING (CH 8 §§4.4, 4.8)

Everything rests on **Ohm's Law: V = I × R** — know any two and you can find the third, and the meter reads each one. The textbook also names the two failures the kids will hunt: an **open circuit** ("a gap… preventing current from flowing") is the dead leg you find with the continuity beep; a **short circuit** (an "unauthorized shortcut… dangerously high current") is why a circuit gets **fused**. The schematic symbols are the shared language for all of it.

HOW THE MEETING RUNS

0:00 · 15 MIN Welcome & safety brief

Open with the standing brief — today it's low-voltage circuits and batteries, plus the iron if you tin a lead. Set the meter before connecting, never short a battery's terminals, and keep irons in their stands. Lower stakes than the soldering bench, but the habit holds.

0:15 · 35 MIN The multimeter

Teach the three readings — voltage, continuity, current — and let every kid take them. The move that matters: match the dial to what you're measuring, leads in the right jacks, and beep out continuity to hunt an open. Tie it back to Ohm's Law ($V = I \times R$). The meter turns "I think it's wired right" into "I know it is."

0:50 · 50 MIN Switches & simple circuits

Start from how a switch makes and breaks a circuit, then have each group build a working circuit on the Simple Circuits Lab Kit: read the schematic, wire it, flip the switch, and the LED lights. No light means an open — walk it with the meter. Every kid builds and measures one before the day is out.

1:50 · 40 MIN Build & measure a circuit

Now make it real for each kid: build the circuit, prove it with the meter (voltage across the LED, continuity of each leg), and read it back against the schematic. This is the rep that turns the symbols and the meter from ideas into skills they own.

2:30 · 20 MIN The Electrical Symbols sheet

Hand out the symbols sheet and walk the ones they'll see all season — wire, switch, battery, resistor, LED, fuse. Three-hole punch it into the notebook today. See the call-out: this free page is the same standard the competition SID is graded against.

COMPETITION DIVIDEND — THE ELECTRICAL SYMBOLS HANDOUT

MATE's free one-page Electrical Symbols handout uses the **same standard the competition SID is graded against**. Print it, three-hole punch it, and keep it in the notebook all year — today's reference becomes April's grading rubric.

2:50 · 10 MIN Wrap-up & homework

Close the loop: have each kid practice reading a simple schematic, keep the symbols sheet in the notebook, and confirm the repo is ready for wiring notes. On the adult track, stage the control-box parts and soldering stations — the next class is a long build.

THE PARENT LANE — SECTION 4.3

Every kid sets the meter, takes the readings, and wires their own circuit — you demonstrate the meter once, then step back. The circuit they build and measure is theirs. Disclose any outside help, including AI, in the documentation.

MATE RESOURCES FOR THIS CLASS

Assign the first three as pre-watch; the symbols handout is kept forever. Source: the TriggerFish ROV Guide.

- Multimeter Basics PRE-WATCH
- How Switches Work PRE-WATCH
- Simple Circuits + Simple Circuits Lab ACTIVITY
- Electrical Symbols handout KEEP FOREVER

COMMON PROBLEMS & QUICK FIXES

SYMPTOM	WHAT'S WRONG & THE FIX
Meter reads nothing / "OL"	Wrong dial setting or leads in the wrong jacks. Match the dial to the measurement; red lead in the V/Ω jack for voltage and continuity.
LED won't light	Polarity backwards or an open circuit. Flip the LED; beep out continuity to find the break.
Switch does nothing	Wired to the wrong terminals. Check the pinout against the symbol.
Readings jump around	Loose lead or a poor contact. Re-seat the leads and the components, then re-measure.

WHAT "DONE" LOOKS LIKE

- Every kid measured voltage and continuity with the meter.
- Every kid built a working circuit and read it against the schematic.
- The Electrical Symbols handout is in every notebook.
- The team can read the six symbols they'll use all season.

ONE SESSION — THIS IS THE CALM BEFORE THE BUILD

This is a measure-and-read class, not a build class, so it fits one session comfortably. The point is fluency with the meter and the symbols — the control box (Class 1.5) is the long build, and it goes far better when the team can already read what it's wiring.